ASIAN INSTITUTE OF MANAGEMENT

Data Mining and Wrangling

Information Retrieval

Session 15 and 16

BSDSBA 2028

04 March 2026

ASIAN INSTITUTE OF MANAGEMENT

Class Administrative Matters

Course Deliverables

Due this Week – R07 (Wed March 18), E4 (Fri March 20), E3S2 (Fri March 20)

Upcoming – ICA07(Wed March 25), E5 (Fri March 27), A1 (Fri March 27),

Mini Project 1 (Mon March 30)

For Checking – R06, ICA04, ICA05, ICA06, E3, Midterm Exam

Session 15 and 16 – Information Retrieval

Gameplan

Session 15

Introduction to Information Retrieval (IR)

Data Preparation for IR

Session 16

Searching by Similarity

IR System Evaluation

Introduction to Information Retrieval

Introduction to Information Retrieval

What is Information Retrieval?

Information Retrieval (IR) is concerned with finding material (usually documents) of unstructured nature (usually text) that satisfies an information need from within large collections (usually stored on computers).

query

data

mining

IR system

collection

1. data mining
2. data mining: concepts and techniques

results

Introduction to Information Retrieval

Information or object retrieval applications

Web Search – searching relevant web pages from a given query.

Text Classification – documents are compared across a fixed set of queries given in advance.

Clustering – finding groupings given a set of documents based on their contents.

Recommender Systems – searching relevant products/objects from a given user/item attributes.

Introduction to Information Retrieval

How do we create IR systems?

1. **Data Preparation** – represent each document or object in a form that is easy to be worked on by algorithms.
2. **Similarity measurement** – define a measure of similarity between documents or objects.
3. **Retrieval and Ranking** – return the most similar objects based on the similarity measure.

Introduction to Information Retrieval

Data Preparation

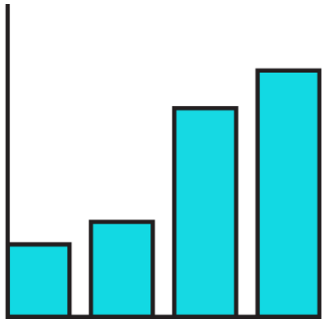
A key step in Information Retrieval, and in most Machine Learning systems for that matter, is ***vectorization*** – the process of transforming raw data (such as text, images, or other signals) into *numerical representations* that machines can efficiently process.

Introduction to Information Retrieval

Data Preparation

Discrete Variables

Values of discrete variables can only take on a set of isolated points.



$$\{x_1, x_2, x_3, \dots, x_n\}$$

Continuous Variables

Values of continuous variables can assume any value within a specified range.

$$\{x \in \mathbb{R} \mid a \leq x \leq b\}$$

Introduction to Information Retrieval

Data Preparation: Discrete Variables

Nominal Data

Data type that can take any arbitrary non-numerical values. No arithmetic information can be applied on these.

Ex. Names, address, gender

Ordinal Data

Data type whose values follow an order.

Ex. T-shirt sizes, likert scale, rankings

Introduction to Information Retrieval

Data Preparation: Continuous Variables

Interval Data

Numerical values where the difference between values is meaningful but there is NO true zero point.

Ex. Temperature (°C or °F), Date, Time, IQ

Ratio Data

Data type that has a natural zero point and supports all the properties of interval data, along with operations of multiplication, division, etc.

Ex. Weight, height, time duration

Introduction to Information Retrieval

Data Preparation: Discrete Variables (Nominal)

One-hot Encoding

The data is converted to binary arrays, where a binary column is created for each possible value of an attribute.

Gender	Is Female?	Is Male?	Is Other?
Female	1	0	0
Male	0	1	0
Other	0	0	1

Data Preparation: Discrete Variables (Ordinal)

Ordinal Encoding

The data is transformed into a numerical value that preserves the natural and inherent order.

Ex. Sizes: Small → 1, Medium → 2, Large → 3

Introduction to Information Retrieval

Data Preparation: Continuous Variables

Normalization

A data preprocessing step done so that features lie in similar range.

Min-Max Scaling

$$x_{min-max} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Standard Scaling

$$z = \frac{(x - \mu)}{std}$$

Robust Scaling

$$x_{robust} = \frac{(x - x_{median})}{IQR}$$

Introduction to Information Retrieval

Data Preparation: Text Data

Text Data Pre-processing Pipeline

1. **Tokenization and Case Folding** – the process of breaking down a sentence or a document into individual components or units called *tokens.* In some cases, the tokens are then converted to have the same case.

2. **Stemming, Lemmatization, Handling Contractions** – the process of extracting the *root word* of a given token.

3. **Stopwords Removal** – the removal of high-frequency words that can be ignored with minimal effect in accuracy.

4. **Word Vectorization** - the process of converting the pre-processed text data into vector form.

Introduction to Information Retrieval

Word Vectorization

Bag-of-Words Vectors

The Bag-of-Words Vector is a vector representation of a text where each component corresponds to the number of occurrences of the word in each document.

doc_id	document
1	mary had a little lamb
2	little lamb
3	as white as snow

BoW Vectors

doc_id	a	as	had	lamb	little	mary	snow	white
1	1	0	1	1	1	1	0	0
2	0	0	0	1	1	0	0	0
3	0	2	0	0	0	0	1	1

Introduction to Information Retrieval

Word Vectorization

TF-IDF Vectors

The **Term Frequency-Inverse Document Frequency** representation attenuates words that are frequently occurring across documents by multiplying the term frequency (from the BoW vector) with the inverse document frequency.

For a given term t, document d, corpus D

$$\mathrm{tf}(t, d) = \frac{\mathrm{count}(t)}{\mathrm{count}(d)}$$

$$\mathrm{idf}(t, D) = \log \frac{\text{number of documents}}{\text{number of documents containing } t}$$

$$\mathrm{tfidf}(t, d, D) = \mathrm{tf}(t, d) \times \mathrm{idf}(t, D)$$

Introduction to Information Retrieval

Similarity Measurement

The L_p-norm

The most common distance function for quantitative data is the L_p-norm given by:

$$Dist(\vec{X}, \vec{Y}) = \left(\sum_{i=1}^{d} |x_i - y_i|^p \right)^{1/p}$$

Two special cases of the L_p-norm is the *Euclidean* $(p = 2)$ and the Manhattan $(p = 1)$ distance.

Introduction to Information Retrieval

Similarity Measurement

Cosine Similarity

The cosine measure computes the angle between two vectors

$$cos(\vec{X}, \vec{Y}) = \frac{\vec{X} \cdot \vec{Y}}{\|\vec{X}\| \|\vec{Y}\|}$$

In this case, the cosine distance between two vectors $\vec{X}$ and $\vec{Y}$ is $1 - \cos(\vec{X}, \vec{Y})$

Introduction to Information Retrieval

Retrieval and Ranking

Once we have our objects and documents defined as vectors and have chosen the similarity measure to use, the retrieval and ranking process is straightforward:

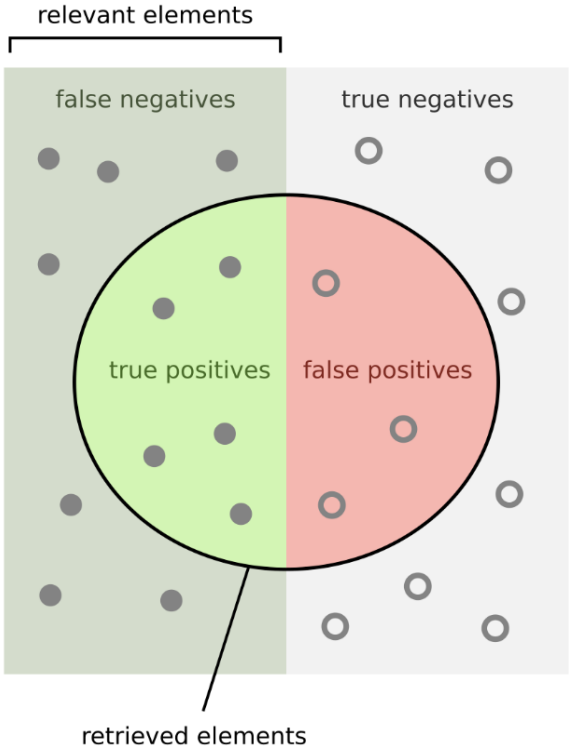
1. **Transform the query** into the same representation as the objects in the database.
2. **Compute the similarity** of the query with each of the objects in the database.
3. **Return the objects** in the database with greatest similarity.

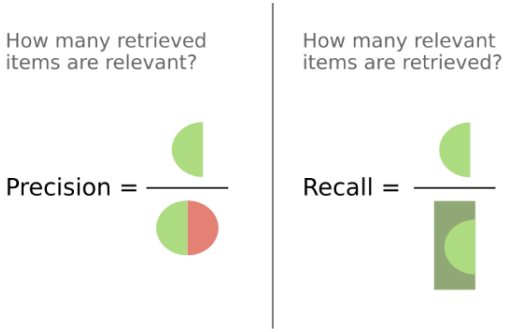
Introduction to Information Retrieval

Evaluation of IR Systems

Recall and Precision

- Recall measures how much of the relevant documents were returned by the IR system
- Precision measures how much of the documents returned by the IR system were relevant.





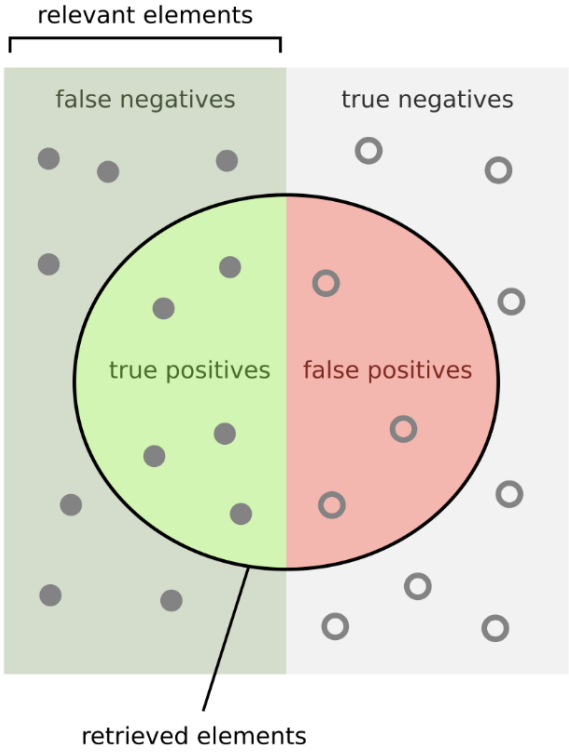
Introduction to Information Retrieval

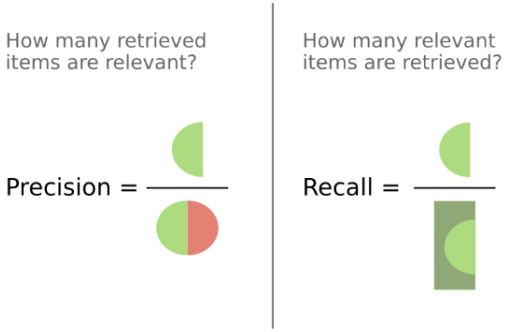
Evaluation of IR Systems

Recall and Precision

$$\text{recall} = \frac{|Rel \cap Res|}{|Rel|} = \frac{TP}{TP + FN}$$

$$\text{precision} = \frac{|Rel \cap Res|}{|Res|} = \frac{TP}{TP + FP}$$





Introduction to Information Retrieval

Worked Example

A student searches in a digital library:

```
“data science”
```

The library has 8 relevant “data science” books, but the IR system suggested the 5 books:

```
1 Introduction to Data Science

2 Python for Data Analysis

3 Cooking for Beginners

4 Machine Learning Basics

5 Travel Guide to Japan
```

Introduction to Information Retrieval

Worked Example

The confusion matrix can be written as:

	Returned	Not Returned
Relevant	3	5
Not Relevant	2	

The library has 8 relevant “data science” books, but the IR system suggested the 5 books:

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1 Introduction to Data Science

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```

Introduction to Information Retrieval

Worked Example

The confusion matrix can be written as:

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$$\text{Recall} = \frac{3}{8}$$

$$\text{Precision} = \frac{3}{5}$$

Introduction to Information Retrieval

Evaluation of IR Systems

Recall and Precision @k

For an IR system that returns the top k results, the precision and recall at k is defined as:

$$\text{recall} = \frac{|Rel \cap Res[1..k]|}{|Rel|}$$

$$\text{precision} = \frac{|Rel \cap Res[1..k]|}{k}$$

Introduction to Information Retrieval

Evaluation of IR Systems

Average Precision

The indication of the effectiveness of an IR system can be measured by the average precision value:

$$AP = \frac{1}{|Rel|} \cdot \sum_{i=1}^{k} \text{relevant}(i) \times \text{Precision@}i$$

The average precision represents the approximation of the area under a recall-precision curve.